A Seed Agent That Improves Its Own State: Guidance and Invention Across Seven Areas, with a Ten-Seed Replication and Preregistered Ablations

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Abstract

This paper studies a narrow form of self-improvement: can one persistent agent improve the state on which a fixed learning procedure operates, and carry that state into settings with different structure? The state is the agent's accumulated experience, the models that guide its search, and the primitives from which it builds candidate solutions. The agent solves rule-inference tasks by searching a catalog of candidates under a scan budget, and it can fail for two reasons: the right candidate lies outside the budgeted portion of the catalog, or its primitives cannot express it. Learned search guidance addresses the first and invention of primitives the second. A change is retained only when a paired interval on held-out worlds is above zero, and a change to the shared primitives must also leave previously learned settings within a stated tolerance. The final protocol, version 11.1, was frozen with file hashes, pre-stated gates and a committed forecast and run on ten sealed random seeds, over seven areas in two task forms. On one held-out draw per seed, never used for acceptance, every area and every pool rose on every seed, with area medians from +32% to +431%; nine of the ten seeds passed all four gates. Seven arms on three of those seeds, under decision rules committed before any arm ran, then measured what each part is worth. Measured as the mean log growth over the seven areas, the agent without learned guidance reached 6% of the full agent's growth and the agent without invention 65%, both lower on every seed. The acceptance test did not earn its cost in growth: keeping every proposal grew as much for 29% of the scans, and keeping every proposal while spending the test's scans on more experience grew more on all three seeds, so the claim that the test drives the growth is withdrawn. What it bought was fewer harmful changes: 2 of 35 kept changes lowered their own area on a fresh set, against 15 of 63 and 17 of 57 under the two cheaper rules. A second evaluation draw, committed before it was scored, gave the same verdict for every arm, with paired intervals. It also bounds the headline growth: the starting state lacks two primitive families by design, and a start with nothing withheld reaches about a third of the full agent's growth before anything is learned. Separately, the agent may change six settings of its own improver, under a test no acceptance used; nine of the ten seeds adopted at least one such change, 13 in all. Twelve came from tests in which the change altered no outcome, and saved nothing over a whole run; one of the ten tests in which it did alter an outcome produced a better state. The paper ends with a table of what is supported, what is withdrawn and what is not shown, and with the limitations found by an audit of its own repository.

1 Introduction

An agent can improve its performance without improving the algorithm that teaches it. It can accumulate useful evidence, learn where to search, or acquire a new primitive that makes a previously inaccessible solution expressible. These are different forms of growth, and they can be carried forward or lost.

This research programme uses *recursive self-improvement* to mean a system producing a successor with a better improvement procedure; that is the definition used here, not a universally accepted one. The present paper studies a narrower question: can one agent improve the persistent state on which a fixed learning procedure operates? That state includes its accumulated experience, the models that guide its search, and the primitives from which it constructs solutions. Changes to this state influence subsequent problem solving and the experience available for later updates, but through version 10 the update procedures themselves do not change; version 11 (Section 5) lets the agent change a fixed menu of its procedure's settings, under the same kind of test. The contribution is the integration of several state-improvement mechanisms within one persistent agent, a replication of its final protocol on ten sealed random seeds, and preregistered ablations that measure what each mechanism is worth. Learned guidance and invention carry the growth. The acceptance test that decides which changes remain does not add growth at the budget studied; it limits harm.

Two levels of improvement. The paper keeps them apart throughout. *Level 0* is improvement of state: experience, models, priors and primitives change, and the code that changes them does not. This is the main subject, and level-0 growth should not be read as recursive self-improvement. *Levels 1 and 2* are improvement of procedure: from version 11 the agent revises its producer's fallback rule from its own ledger and tests changes to six settings of the improver on separate branches. Those results are reported separately, and Section 9 states how small they are.

Retaining improvements is straightforward only when their effects remain local. In this system the primitives are shared across settings. An update that helps the current setting can therefore change the search space of a setting learned earlier. The experiments expose exactly this failure and motivate testing shared changes wherever they will be used.

The agent is called a *seed* because it is meant to be the thing that grows: a saved state plus fixed update procedures, run for successive cycles of experience. To avoid a clash with pseudorandom seeds, the paper says *seed agent* or *agent instance* for the artifact and *random seed* for identifiers such as 644101.

Antecedents. Each form of growth had been shown separately in this programme, in separate code. Within-life accumulation: an agent that keeps the atoms of whatever it submits, right or wrong, raised its own solve rate four to ten times across a hundred tasks at every scan budget from 128 to 4,096, and being told whether it was right added nothing. Learned search guidance: a grouping of primitives learned without labels from 1,600 worlds solved 6,993 and 8,077 more tasks over 18 budget cells than the same agent on a grouping from 400 worlds, on prespecified random seeds, audited. Expanded representational reach: in a string-manipulation language, a system that synthesised primitives from its failures solved 20 of 24 tasks where its fixed grammar solved 5, on two random seeds. None of these was recursive in the programme's sense; every attempt at a better child improver returned a null result. What they shared was a state that grew with experience and a learner that did not. These antecedents are unpublished; their reports are in the repository, indexed by `GROWTH-DISTILLATION-2026-10-10.md`.

2 The task, and the words this paper uses

2.1 A task in this system

Each task asks the agent to infer an unknown Boolean rule over characters from a small set of visible examples. The agent searches a catalog of candidate rules and submits one that agrees with

the visible examples. Hidden examples are reserved for evaluation. The search budget limits how many candidate rules the agent can inspect.

Here is one task from the implementation, drawn from a world whose hidden structure is described below. Characters are given by their code within a 256-character alphabet. The true rule is that bit 7 of the code is 0, that is, the code is below 128. Each visible example is a string whose longest prefix of characters satisfying the rule is the answer; the first three of the eight visible examples are

answer prefix	rest of the string
119, 115, 48, 47, 121	152, 242, 205
23, 10, 101	168, 131, 144, 143, 137
61, 7, 118	240, 178, 187, 203

A candidate rule is consistent with an example when every character of the answer prefix satisfies it and the first character after the prefix does not. The agent's catalog at the start of this work has 82,758 candidate rules. Of these, 77 are consistent with all eight visible examples, and 17 of the 77 also reproduce the sixteen hidden examples. Scanning the catalog in its starting order with a budget of 1,024 rules, the agent meets exactly one consistent candidate, "code below 144", which is built from the single primitive `threshold:144` and its complement. It submits that rule. The rule fails a hidden example, so the task is not solved; the agent is not told this; and the primitive `threshold:144` enters its experience of this world as if it belonged there. The right rule, built from the primitive `bit:7:0`, was in the catalog all along but outside the 1,024 rules inspected. The agent's record of this submission notes that it found one consistent candidate in its window, which makes the submission a *confident pick* in the sense defined below; on average such picks are correct 90% of the time, and this one was not.

2.2 Two hierarchies: how problems are generated, and how experience is scheduled

Two organisations should be kept apart. On the generation side, an *area* is a distribution over worlds with a particular hidden structure of primitive co-occurrence; a *world* is one hidden set of 14 primitives, and every task of that world is drawn from a rule built only from them, so a world's tasks are related. On the schedule side, a *life* is one cycle of experience and update: the agent lives 48 worlds of one area, 100 tasks each, and only then proposes and tests changes to its state. The task above came from a world of the first area, drawn from the families *bit*, *divmod* and *gcd*.

2.3 Terms

Term	What the reader should understand by it
seed agent	A persistent agent consisting of saved state and fixed update procedures.
atom	An elementary Boolean predicate over characters, such as "bit 7 of the code is 0", used as a building block for candidate rules. There are 313 in 9 families named for their arithmetic operation.
language	The collection of atoms the agent currently holds; not a natural-language vocabulary.
hypothesis catalog	Every atom, every pair of atoms from different families under and, or and xor, and every complement, reduced to one row per distinct behaviour over the alphabet. Rebuilt from the language whenever it changes: 82,758 rows from the starting 260 atoms, 166,596 from all 313.
world	A hidden set of 14 atoms from which related tasks are generated.
area	A distribution over worlds with a particular co-occurrence structure.
life	One experience-collection and update cycle: 48 worlds, then proposals and tests.
earned experience	The atoms behind the agent's own submissions in a world, kept whether the submission was right or wrong.
confident pick	A submission made when exactly one consistent candidate lay in the scanned window. Not uniqueness in the whole catalog, and not certainty.
world and self model	An area's model of which atoms occur together and how often the agent finds each; it guides search.
carried state	Everything the agent keeps between lives.
prove	Run an empirical acceptance comparison between the carried state and a candidate on held-out worlds; not a mathematical proof. The paper says *test* and *accept*.

3 Why the agent fails, and what it does about it

The agent can fail for two different reasons. A suitable predicate may already exist in its catalog but lie outside the portion inspected under the search budget, as in the task above. Alternatively, the current language may lack the primitives needed to construct it; the starting language omits two of the nine families, so a third of the rules in some worlds cannot be expressed at all. Learning a model can help with the first problem by changing which hypotheses are searched first. Extending the language can help with the second by adding new building blocks. Pruning addresses a related cost: added primitives enlarge the catalog and can dilute the budgeted search without improving performance. Two qualifications prevent misreadings. A task left without any consistent candidate is not proof that a primitive is missing; it means the budgeted search found none. And invention happens within a fixed expression grammar, not by unrestricted construction of new algorithms.

3.1 What persists

The carried state is the language; the experience, one record per world lived, with the area and the atoms behind the agent's confident picks and behind all its picks, by atom name so that the records survive a change of language; per area, a world and self model and a flag saying whether to build a search prior from that area's experience; the invented atoms and the life they entered; a ledger of every candidate change with its test; and the resource spent on living, testing and inventing. The language, the experience and the ledger are shared across areas; the model and the prior belong to an area.

3.2 The operations

Live collects evidence while solving related tasks. The agent draws fresh worlds of an area and lives each as a stream of 100 tasks at the scan budget. Within a world, the atoms of each submission reorder the catalog for the next task, which is the within-life accumulation of the antecedents. Two carried objects reorder it further. The area's model, when it exists, turns the atoms earned so far in the current world into beliefs about which groups of atoms the world uses, and those beliefs promote the rows those groups build. The area's prior, when it exists, starts the scan from an order given by how often atoms co-occurred in the agent's own earlier worlds there, with half of each task's budget reserved for the order it would use with no prior at all. With neither, the agent scans the catalog in one fixed order, since version 26 a hash of each row's content. The agent receives no correctness feedback on any submission.

Reflect uses accumulated evidence to improve future search guidance. The world and self model describes both which primitives tend to occur together and how often the agent discovers particular primitives; its purpose is to use the agent's own submissions to guide subsequent search. Latent: a group label per atom, and which groups each world used. Observed: the confident picks of every world of the area. Fitted, by expectation-maximization: the labels, how often each set of groups occurs, and for each atom a *discovery rate*, how often the agent finds the atom when it is present in a world, and an *import rate*, how often the agent earns the atom through a wrong pick in a world where it is absent. An atom whose discovery rate falls below 0.10 is filed in no group: the agent meets it only through its own mistakes. Downstream, during a new world the collected atoms update the agent's beliefs about which groups are present, and those beliefs promote the corresponding hypotheses in the search order. The model's size, 6 to 8 groups and 2 to 4 groups per world, is chosen by the likelihood of the agent's last fifth of worlds in that area. The refit model and a prior from the area's experience form one candidate change. The fuller account of the model is in the companion paper.

Extend proposes additional primitives from uncovered tasks. For the third of the life's worlds with the most tasks left without any consistent candidate, and only worlds with at least 15 such tasks, the agent enumerates expressions of depth one from the ordinal grammar and tests each against the constraints of the uncovered tasks, alone, complemented, or joined by and, or or xor to one atom it earned in that world. A candidate atom is proposed only if it fits at least two uncovered tasks: an atom a world is built from recurs across the world's rules, and a chance fit does not. The grammar yields 3,456 candidates distinct up to complement, searched smallest first; at most 40 atoms are proposed per life.

Prune proposes removing unused inventions. An invented atom is proposed for removal two or more lives after it entered if it has never appeared in a confident pick, or if every area model that has seen it rates it as found only through wrong picks.

Prove determines whether a change deserves to persist. After a life the agent proposes all three kinds of change at once and tests the combination; if the combination fails, it tests each part alone. A test runs the current state and the candidate on a held-out set of the area's worlds, drawn for that life and never lived. Under the rule in force at the end of this work, the candidate is accepted only when its gain over the current state, paired over the held-out worlds, has a 95% bootstrap interval above zero; a candidate clearly behind on the first half of the set is stopped there. A candidate that changes the language is also run on a smaller held-out set of each other

area the agent has lived in and is rejected if its total there falls more than 2% below the current state's, a relative decrease on that set. Every held-out lifetime is charged to the agent's resource. The agent therefore receives aggregate performance feedback when selecting updates, though never per-submission feedback while collecting experience. Section 5 gives the history of this rule, which was revised four times during development (Table 2), by the experimenter and not by the agent.

Mutate tests changes to the agent's own procedure (version 11). Six settings govern the producer and the rule that revises it: patience and probation (how many rejections retire a kind of proof, and for how long), the extension cap, the recurrence an invented atom needs, the share of hard worlds invention looks at, and the discovery rate below which pruning proposes an atom. Each task form keeps its own values. Every second life of the agent, the ledger votes on which setting to move and which way (a kept extension that filled the cap votes for a larger cap, for example), and the agent forks: branch a proposes and proves under the current settings, branch b under the mutated one, from the same lived worlds, for one life, or two for patience and probation, which act across lives. Both branches are scored on a meta set of 48 worlds drawn for that test and used by no proof. The mutation is adopted, with branch b's state, if the paired interval is above zero, or if the mean is not negative and branch b's proof cost is at least 10% lower; otherwise branch a continues. Meta-tests are charged to the resource and capped at 30% of proof scans. The losing branch is snapshotted so the auditor can replay the decision.

Figure 1 shows the cycle and what travels from one cycle to the next. The procedures in the boxes never change; the state on the return path does.

4 Areas

Mechanism worlds (`mech`). A world draws three of seven families and its 14 atoms from their pooled atoms. The starting language lacks the two remaining families, popcount and xormod, so a world that uses one of them holds rules the agent cannot express. A fifth of the viable family triples are withheld from the agent's lives as *untrained combinations*. Lives draw three quarters of their worlds from trained triples and a quarter from triples with an unfamiliar family; held-out sets hold equal numbers of trained, untrained and unfamiliar worlds.

Drawn partitions (`drawn7`, `drawn11`, `drawn13`). Every atom of every family is assigned at random to one of a few groups; a world picks some groups and draws its 14 atoms from their union, and each atom slips to a uniformly random atom with some probability. The partition cuts across families, so nothing learned about mechanism worlds applies except the language. The three areas differ in structure: 9 groups with 3 per world and slip 0.15; 12 groups with 4 per world and no slip; 6 groups with 2 per world and slip 0.15. The agent is not told how many groups a world uses.

Metric worlds (`metric777001`). Each atom has a behavioural descriptor, the normalised Fourier and Walsh spectra of its truth table over the alphabet. A sparse weighting over descriptor dimensions is drawn once and shown to nobody. A world is a centre atom together with the atoms nearest it under that weighting, fourteen in all, the centre included as the nearest to itself. A quarter of each life's worlds are centred on an atom of the two families the starting language lacks. Here the hidden structure is behavioural rather than syntactic.

String tasks (`str7`). A second task form, added for the eighth confirmation version. A task is three visible and six hidden input-output pairs of a hidden program in the trapdoor string language

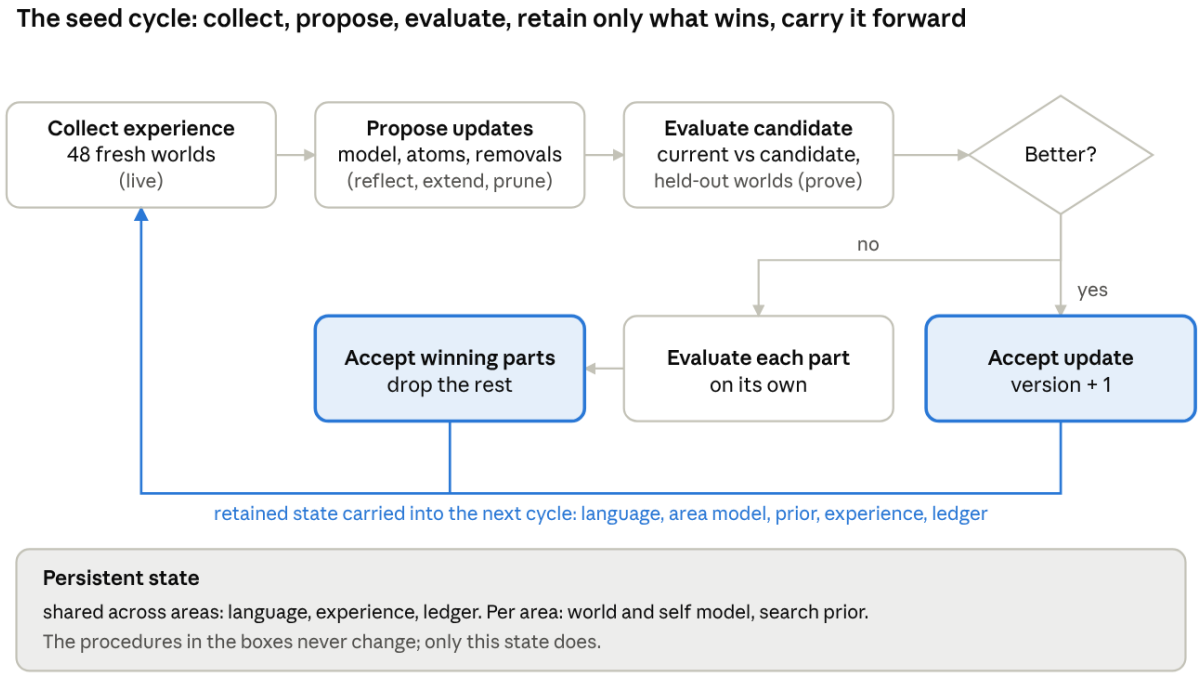


Figure 1: One cycle of the seed agent. It collects experience in 48 fresh worlds, proposes updates of three kinds, evaluates the combined candidate against the current state on held-out worlds drawn for that cycle, accepts it if it wins and otherwise evaluates each part on its own, and carries the retained state, language, area model, prior, experience and ledger, into the next cycle. The language, experience and ledger are shared across areas; the model and prior belong to an area. Implementation names in parentheses.

of [18]: a program concatenates up to three rows, each the substring between two positions, with or without a case operation, or a constant. The positions (character offsets, and the start or end of the first, second or third run of letters, digits, upper-case or lower-case characters), the three case operations and six constants are the keys, 58 in all, and the 9,400 rows they build are the catalog. A world is 14 keys from three of seven key families and 100 tasks over programs on those keys alone, the mechanism structure transposed, with trained, untrained and unfamiliar pools; the twelve positions at upper-case and lower-case runs are withheld from the starting language. The agent's window is the first b rows of the catalog in the order its evidence, prior and model give; a task is solved by the first program within the window that reproduces the visible pairs, and counted only if it reproduces the hidden ones. Extend proposes keys the language lacks that complete programs for at least two uncovered tasks of a stuck world. The same state, operations and acceptance test apply; a language change is checked across areas of the same form only, since the two forms build their catalogs from disjoint keys.

What a random seed varies, and what it does not. The three drawn partitions and the two metric weightings are fixed by the area's name: every random seed meets the same five structures, and the mechanism and string areas have one structure each. A random seed varies the worlds drawn, the held-out sets and, through them, the agent's trajectory. Results over seeds are therefore conditional on these seven structures; they say nothing about a new partition or weighting. The

metric areas are also small. A metric world is fixed by its centre, and only 118 and 80 centres yield a valid world under the two weightings, so an agent that lives 288 worlds of the first and 192 of the second meets most of them (Section 6).

5 Evidence boundaries, protocol and chronology

What information is used where. Table 1 lists the evidence each stage reads and the decision it supports. The model-selection subset is part of accumulated experience, not a separate dataset; the final evaluation sets are separate and were never used to accept anything.

Table 1: Evidence and decisions by stage.

Stage	Evidence or measurement used	Decision it supports
Task experience	Visible examples, submissions, consistency counts, uncovered tasks	Search and experience collection; no per-submission correctness feedback
Model selection	Held-out likelihood on the last fifth of the area's accumulated worlds	Choice among model sizes
Update acceptance	Aggregate held-out solve counts, paired over worlds	Whether a proposed state change is retained
Final evaluation	Further held-out sets not used to accept updates	The reported gains

Schedule of the development agents. This paragraph and the next describe the three development agents; the settings of the final protocol follow them. A life is 48 worlds of 100 tasks at a budget of 1,024 scans per task. State is saved after every operation, so lives in different areas can be interleaved and a run resumed. The main agent, `s1` on random seed 644101, lived twelve lives in mechanism worlds, four in `drawn7`, four in `metric777001`, three in `drawn11` and two in `drawn13`, in that order except that mechanism lives 10 and 11 came last. Agent `s2` on random seed 643101 lived three mechanism lives, then two in `drawn7` and two in `metric777001`; agent `s3` on 645101 lived the same schedule.

Held-out sets. An acceptance test uses 24 worlds per pool of the area (72 for mechanism worlds, 24 for a single-pool area), and the cross-area check 12 per pool. From version 11 of `s1` onward every life draws its own held-out set; the first eight mechanism lives (versions 0 to 10) reused one fixed set, and the numbers read on it carry the optimism that reuse allows. The reported gains come from one further held-out set per area, drawn with a salt no test used, on which the carried state and the starting state (starting language, no model, no prior) are run once each.

The final protocol (version 11.1). Every sealed seed of the final protocol runs `confirm_seed.sh`: 21 lives, in the order four in mechanism worlds, three in `drawn7`, three in `metric777001`, two each in `drawn11`, `drawn13` and `metric555001`, three in `str7` and two more in mechanism worlds; 96 worlds per life at 1,024 scans per task; an acceptance set of 48 worlds spread over the area's pools, kept until a change is accepted on it and then drawn again; a stop at the half set when the candidate is behind; one extension set of the same size when the interval straddles zero and a doubled set could clear it; for a language change, a cross-area set of 12 worlds per pool, and at least 24 worlds, in each other area of the same task form, with a tolerance of 2%; a meta-test every second life on a set of 48 worlds, over one life or two, capped at 30% of testing scans; and four worker processes. The

final check draws 24 worlds per pool with a salt nothing else uses. The arms below use the same schedule and sizes with the meta-level off.

The acceptance rule, by version. The rule was revised by the experimenter as results exposed its weaknesses (Table 2); the agent did not alter it. The ledger records which rule accepted each change.

Table 2: Which acceptance rule governed which versions of `s1`.

Versions (lives)	Rule
0 to 10 (mechanism lives 0 to 7)	Candidate accepted if it solves strictly more than the current state on one fixed held-out set; single-area test; one model for the one area.
11 to 22 (mechanism 8 to 9, `drawn7` 0 to 3, `metric` 0 to 3)	As above, but the held-out set is drawn fresh for each life and models and priors are kept per area.
23 to 25 (`drawn11` 0 to 2, `drawn13` 0 to 1, mechanism 10)	Combination tested first, then parts; a language change must also stay within 2% on every other area lived; groups per world chosen by held-out likelihood.
26 (mechanism 11)	Accepted only if the paired 95% bootstrap interval of the gain is above zero; tie-break a hash of row content; pruning also on the self model's verdict.
after 26	Half-set early stop, version snapshots for audit, held-out sets of at least 48 worlds and cross-area sets of at least 24 worlds; the rule of the confirmation runs.

What the agent never sees. A family label, a true world, which pool a world came from, or whether a submission was correct. Labels enter only in scoring, to say which pool a world came from and which family an invented atom reproduces.

Resource. The ledger charges living (worlds $\times$ 100 tasks $\times$ budget scans), every held-out lifetime of every test, and every invention test.

6 Results

Did performance improve? In ten of eleven agent-and-area pairs of the development runs, yes, in all 89 pairs of the first nineteen sealed seeds, and in all 70 pairs of the ten seeds of the final protocol. At a fixed budget of 1,024 scans per task, the main agent's mechanism-world solve rate on held-out worlds never used for acceptance rose from 17.1% to 28.8% over twelve lives (1,232 to 2,072 of 7,200 tasks), with one qualification: trained family combinations rose from 26 to 49 tasks per world and worlds with an unfamiliar family from 5 to 19, but untrained family combinations fell from 20.5 to 18.6 (Section 8). Figure 2 carries the overview and Table 3 the counts. Five-area coverage is the main agent's; the two further agent instances cover mechanism worlds for three lives and the first drawn partition and metric worlds for two lives each. The exception is `s3` in the drawn partition, where nothing was accepted in two lives on 24-world held-out sets and the retained state reads 280 to 246 on the untouched set: the loss came from an extension of 40 atoms accepted in metric worlds, which passed its 12-world cross-area check on the drawn partition (124 to 148) and failed on the larger untouched set. The cross-area check is too small.

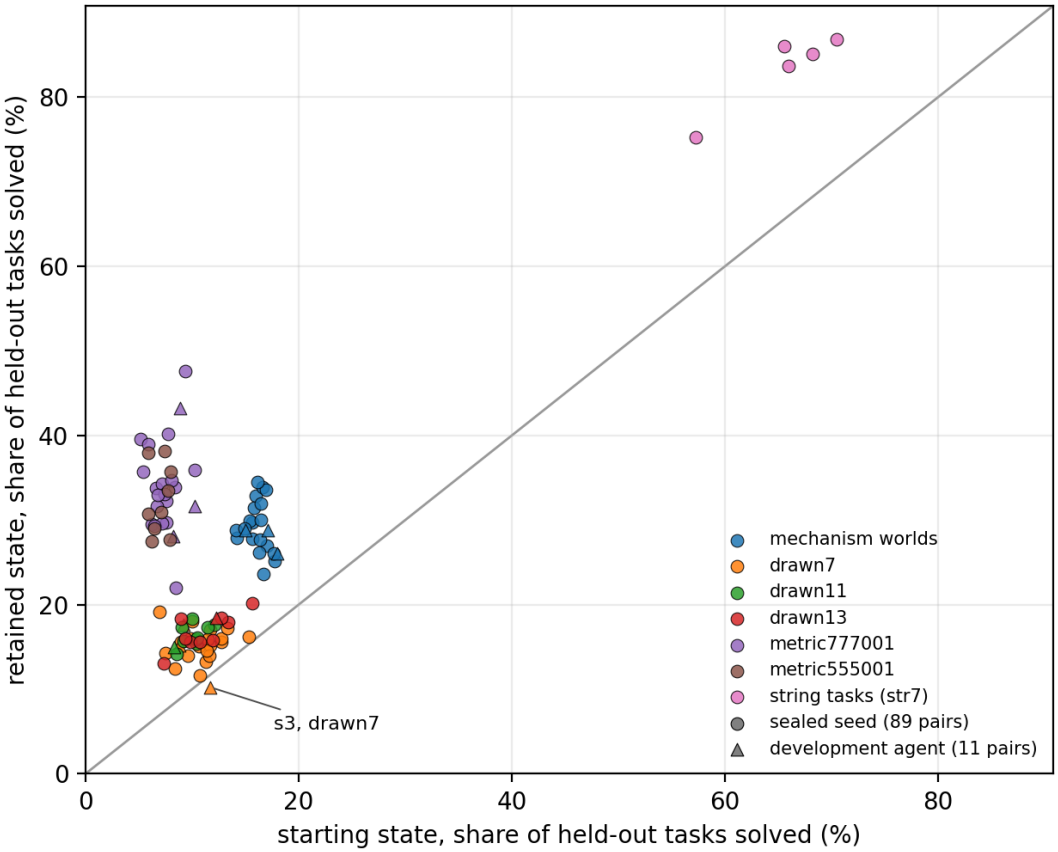


Figure 2: Share of held-out tasks solved at budget 1,024 on sets no acceptance test used: the starting state against the retained state, one point per agent instance and area, 100 pairs in all (the three development agents in eleven, the nineteen sealed seeds in 89). A point above the diagonal improved. Each point is one draw of 24 worlds per pool; the check recorded totals only, so no interval is attached (Section 9). One pair fell, `s3` in the drawn partition.

Which updates were useful? Model updates were the most consistently accepted kind of change: on the main agent a refit model was accepted on every mechanism life from 0 to 6 and on the first tested life of every other area, and on `s2` and `s3` on all three mechanism lives. Language extensions were accepted less often, 6 of 24 proposed on the main agent, and every one proposed on `s2` and `s3` in mechanism worlds was rejected. Accepted removals, four on the main agent, took out 23, 26, 25 and 55 invented atoms. Over 26 versions the main agent accepted 16 models, 6 extensions and 4 removals and rejected 18 extensions, 7 models and 1 removal. These are acceptance outcomes, not a decomposition of the final gain; the arms below give the decomposition.

What was invented? At version 26 the language holds 111 invented atoms: 26 reproduce true popcount atoms and 29 true xormod atoms, the two withheld families, and 56 are behaviourally new. Invented atoms entered only through an accepted test. In metric worlds a model fitted from the first 48 worlds was alone worth 975 tasks on that life's held-out set, and in the drawn areas the agent chose four groups per world for the 12-group area, its true structure, and four for the 6-group area, where the truth is two.

Table 3: Tasks solved on held-out sets never used by an acceptance test, 24 worlds per pool at budget 1,024, all under the content-stable tie-break.

Agent (random seed)	Area (lives)	Start	Retained	Tasks
s1 (644101)	mechanism worlds (12)	1,232	2,072	7,200
s1	drawn7, 9 groups of 3 (4)	211	364	2,400
s1	metric worlds (4)	427	2,076	4,800
s1	drawn11, 12 groups of 4 (3)	199	360	2,400
s1	drawn13, 6 groups of 2 (2)	294	442	2,400
s2 (643101)	mechanism worlds (3)	1,292	1,879	7,200
s2	drawn7 (2)	221	424	2,400
s2	metric worlds (2)	492	1,520	4,800
s3 (645101)	mechanism worlds (3)	1,081	2,074	7,200
s3	drawn7 (2)	280	246	2,400
s3	metric worlds (2)	394	1,349	4,800

What failed? Two things, each of which changed the acceptance rule. A change accepted in one area damaged a learned area (Section 7). And the grown language, through atoms the agent's own model rated as never found, diluted the budgeted search on untrained combinations (Section 8). A third weakness was in the rule itself: on life 10 a model was accepted for a gain of 24 tasks on 72 worlds that read 82 lower on the untouched set, which is why acceptance now requires an interval above zero.

What did it cost? By version 26 the main agent had spent 122.9 million scans living, 302.3 million testing, and 831,738 invention tests. Testing cost two and a half times living, because each life tested up to four candidates on 72 worlds against a freshly scored current state and the cross-area checks added to that. The per-life held-out totals the agent decided by are in the ledger; they lie within a few percent of the untouched-set numbers except for the life-10 case above.

Confirmation on sealed random seeds. After the design was frozen, with the hashes of its source files, each protocol version was run on random seeds never used before, against pre-stated gates and a committed forecast: growth of at least 15% in every area, no pool of any area below 90% of its start, testing within a stated multiple of the cost of living, and every accepted change replaying exactly. Seven of the eleven revisions followed a failed gate; four (versions 5, 8, 9 and 11) extended or repaired the design after a pass. Tables 9 and 10 give each seed's outcome, and Appendix A gives the chronology. No replay overturned an acceptance.

The final protocol on ten seeds. A reviewer of an earlier draft pointed out that the sealed seeds spanned ten protocol versions, with one completed run of the final one. Version 11.1 was therefore run unchanged on eight more sealed seeds, 919127 to 919134, which with 919121 and 919122 makes ten under one set of file hashes (Tables 4 and 5). Every area and every pool rose on every seed. Nine seeds passed gates 1 to 4; seed 919127 failed the growth gate in `drawn13`, 328 to 343, a rise of 5% against the 15% required. The forecast committed before the first of the eight opened gave 0.2 to all eight passing, and seven did. Seeds 919121 and 919122 are the two on which version 11.1 was first run and kept, so the figure for seeds that no design decision depended on is seven of eight; seeds 919123 to 919126 were reserved for later drafts and never run. The two pools

Table 4: Version 11.1 on ten sealed seeds: change from the starting state to the retained state on fresh held-out sets, by area and pool. Median and minimum over seeds, 95% bootstrap interval of the median (1,000,000 resamples of seeds), and the number of seeds below the start. Each seed contributes one draw of 24 worlds per pool; the interval is over random seeds for the fixed area structures.

Area or pool	Median	Weakest seed	95% interval	Seeds below start
mechanism worlds	+84%	+64%	+77% to +90%	0
trained	+51%	+22%	+43% to +58%	0
unfamiliar	+410%	+165%	+347% to +585%	0
untrained	+67%	+13%	+45% to +91%	0
drawn7	+61%	+21%	+49% to +91%	0
metric777001	+431%	+209%	+264% to +511%	0
seen centre	+311%	+147%	+198% to +358%	0
unfamiliar centre	+802%	+406%	+478% to +1075%	0
drawn11	+48%	+18%	+24% to +75%	0
drawn13	+66%	+5%	+51% to +92%	0
metric555001	+306%	+137%	+269% to +359%	0
seen centre	+222%	+89%	+187% to +268%	0
unfamiliar centre	+652%	+394%	+507% to +1678%	0
str7	+32%	+20%	+29% to +35%	0
trained	+21%	+10%	+16% to +26%	0
unfamiliar	+66%	+49%	+55% to +69%	0
untrained	+19%	+8%	+15% to +23%	0

named in advance as the ones to watch, the untrained pools of the mechanism and string areas, rose by at least 13% and 8%, with medians of 67% and 19%. Level-0 proofs cost 2.10 to 3.04 times living and all proofs 2.56 to 4.15, inside the limits of 3.5 and 5.0 on every seed. All 123 accepted changes replayed exactly on the platform that ran them, as did every policy revision and meta-proof; Section 13 says what a replay covers. Each ratio in Table 4 is one draw of 24 worlds per pool, and the intervals are over random seeds for the seven fixed area structures.

Nine of the ten seeds adopted at least one change to the procedure, 13 in all. Seven of the 13 raised the extension cap to 80. Twelve were adopted under the cheaper branch of the rule, a score no lower at a tenth less cost, and one, a recurrence of 3 on seed 919122, because it produced a better state. Seed 919127 adopted nothing.

These adoptions need their denominators. Forty-six meta-tests ran on the ten seeds. In 36 of them the two branches scored the same on every world of the meta set, and all twelve cheaper adoptions are among those 36: the changed setting altered no outcome on the test, and the saving of a tenth was measured over the one or two lives of the branch. Over a whole run the saving did not hold. The full arm of the next paragraphs is the same agent with the meta-level off, on three of these seeds. Seed 919127 adopted nothing, and its level-0 testing cost is identical with the meta-level on or off. Seed 919128 adopted a patience of 2 and its level-0 testing cost 0.5% less than with the meta-level off, for the same kept changes. Seed 919129 adopted an extension cap of 80 and its level-0 testing cost 18.5% more. The meta-tests themselves added 32% to 43% to the testing bill on the three seeds. The one better adoption is one of the ten tests whose branches differed; a rule that passes at a one-sided 2.5% level is expected to pass a quarter of a test by chance among ten, so this result needs replication before it counts for much.

Table 5: Version 11.1 per seed: cost of level-0 proofs and of all proofs relative to living (limits 3.5 and 5.0), accepted changes, all of which replayed, procedure changes adopted, and the gate outcome.

Seed	Level-0	All proofs	Accepted	Adopted	Gates
919121	2.10×	2.56×	11	1	all pass
919122	2.71×	3.67×	14	3	all pass
919127	2.61×	3.63×	12	0	growth fails
919128	2.30×	3.05×	10	1	all pass
919129	2.97×	4.04×	13	1	all pass
919130	2.35×	3.43×	11	2	all pass
919131	2.76×	3.52×	16	1	all pass
919132	2.26×	3.02×	10	1	all pass
919133	3.04×	4.15×	13	2	all pass
919134	2.51×	3.68×	13	1	all pass

Table 6: Recorded full-set proofs of the sealed seeds judged again on part of their own set: full-set interval passes the smaller set would lose (a pass is kept only if it also clears the cross-area check) and full-set interval failures it would pass.

Set size	Passes lost	Failures passed
half	125 of 435 (29%)	30 of 467 (6%)
quarter	207 of 435 (48%)	37 of 467 (8%)

Set size and tolerance. How much of the acceptance test's verdict depends on its size can be read offline from the 1,517 recorded proofs of the sealed seeds, of which 902 ran the full set without an extension. Table 6 judges those again on half and a quarter of the same worlds: the smaller sets save half and three quarters of the test's runs, lose 29% and 48% of the full set's 435 interval passes (280 of which were kept; the cross-area check refused the rest), and pass only 6% and 8% of its interval failures. Reducing the set mostly costs power and rarely admits what the full set refused. A candidate that was never kept has no later fresh-set score, so the reference here is the full-set verdict, not the truth. The cross-area tolerance matters more than its small size suggests: of the 238 candidates that reached a cross-area check, 49 would pass at a tolerance of 0%, 80 at the 2% used, 152 at 5% and 211 at 10%.

What each part is worth: preregistered arms. A reviewer asked for ablations and for a test of whether the acceptance test earns its cost. Seven arms ran the version 11.1 schedule with the meta-level off, on three of the ten seeds (919127 to 919129), every arm from the same starting state on a seed. *Full* is the frozen agent. *Accept-all* keeps every proposed change and runs no test. *Point* keeps a change whose point gain on half the held-out set is positive, with no interval, extension set or cross-area check. *Matched* keeps every change and scales the schedule so that its living scans equal full's living plus testing scans on that seed. *No-guidance* never fits a model or a prior; *frequency* replaces the model with a prior from each atom's earned frequency; *no-invention* never extends or prunes the language. The measure is G, the mean over the seven areas of ln(retained/start) on the fresh held-out sets. The arms, four decision rules and a forecast were committed before the first arm ran, in the commit tagged `prereg-review002-arms`. That commit was local: the arms finished before it was pushed, so its timing rests on the local commit date, the minute the first run started. Three seeds are too few for intervals, so the rules read the per-seed sign (Table 7).

Table 7: The review-002 arms on seeds 919127 to 919129, meta-level off. G is the mean over the seven areas of ln(retained/start) on fresh held-out sets; the difference from full is per seed; scans are live plus proof, mean over seeds, in millions; harmful counts kept changes that lowered their own area on a fresh set. Outcomes follow the rules committed before any arm ran; invention is judged on the unfamiliar-family pools of mech and str7, not on G. Brier score of the eight committed forecasts 0.126.

Arm	G	G − full, per seed	Scans	Kept	Harmful	Outcome
full	0.695	–	717	12, 10, 13	2 of 35	–
accept-all	0.691	+0.045, −0.027, −0.030	206	21, 21, 21	15 of 63	rule 3: inconclusive
point	0.706	+0.066, −0.023, −0.009	366	18, 21, 18	17 of 57	rule 3: inconclusive
matched	0.791	+0.150, +0.059, +0.080	721	76, 70, 74	–	rule 2: withdraw
no-guidance	0.045	−0.650, −0.625, −0.676	341	0, 1, 0	–	rule 4: shown
frequency	0.513	−0.128, −0.184, −0.235	666	9, 10, 6	–	rule 4: shown
no-invention	0.450	−0.181, −0.290, −0.265	431	11, 9, 9	–	rule 4: shown

Guidance and invention carry the growth. Full beat no-guidance, frequency and no-invention on all three seeds, which is the preregistered rule 4 for each. Without a model or a prior the agent reached a G of 0.045 against full's 0.695, 6% of the growth, and kept one change in three runs. A frequency prior in place of the model reached 0.513, but 88% of that gap lies in the two metric areas; over the other five areas full led by 0.001 to 0.083. Invention was judged on the unfamiliar-family pools, which the starting language cannot fully express: without it the mechanism pool ended at 7.4 to 11.8 tasks per world against full's 17.2 to 22.7, and the string pool at 59.1 to 61.7 against 81.9 to 86.7. In G the agent without invention reached 65% of full's growth, losing most in the metric areas and `drawn13`; in mechanism worlds it was slightly ahead on two seeds.

The acceptance test buys safety, not growth. Against the cost arms the preregistered rules give: accept-all and point are inconclusive (rule 3), each ahead of full on one seed of three, and matched is ahead on all three seeds, by 0.150, 0.059 and 0.080 (rule 2). Rule 2 obliges this paper to withdraw the claim that the acceptance test is its central contribution and to say that the test did not earn its cost at this budget and on these areas; it does both. Accept-all matched full's growth for 29% of its scans, because testing cost about 2.5 times living. Most of matched's lead also lies in the two metric areas, which have few centre worlds per set and draw different worlds on another platform (Section 13); over the other five areas it led by 0.015. What the test did buy was fewer harmful acceptances, counted as kept changes that lowered their own area on a fresh set of 12 worlds per pool: 2 of 35 for full, 15 of 63 for accept-all and 17 of 57 for point. These are raw counts with no significance test, and matched, with about four times accept-all's acceptances, was not counted. Nor is the test a guarantee. The one change the no-guidance arm kept, an extension of 35 atoms in `metric555001` accepted on an interval of 3.3 to 10.7 tasks per world, passed the 2% cross-area check (mechanism worlds 717 to 711) and then cost mechanism worlds 13.5% and `drawn7` 19% on the fresh sets. The forecast committed with the rules has a Brier score of 0.126 over its eight events, against 0.25 for always saying one half; its largest miss was matched leading on two or more seeds, given 0.45.

What the arms leave open. Four qualifications belong beside these verdicts. First, matched is matched on scans and on nothing else. Scans do not charge the model refits or the invention tests, and matched's lives took 2.5 times full's wall-clock time (5,577 seconds against 2,198, mean

Table 8: The arms scored again on a second evaluation draw (salt 20001, 24 worlds per pool) that nothing had used. G as in Table 7, mean over the three seeds, on the first and the second draw; then the difference from full on the second draw, per seed. A star marks a difference whose paired 95% bootstrap interval over worlds (10,000 resamples within pools) excludes zero. The last row is the starting state given every generator atom, with no model and no prior.

Arm	G, first draw	G, second draw	G − full on the second draw, per seed
full	0.695	0.763	–
accept-all	0.691	0.771	−0.003, +0.018, +0.006
point	0.706	0.771	+0.002, +0.023, −0.003
matched	0.791	0.879	+0.139*, +0.125*, +0.082*
no-guidance	0.045	0.057	−0.737*, −0.561*, −0.821*
no-invention	0.450	0.469	−0.280*, −0.300*, −0.304*
frequency	0.513	0.569	−0.116*, −0.192*, −0.275*
start, nothing withheld	–	0.271	–

over seeds) and ran 3.4 times its invention tests; accept-all, at 29% of full's scans, took 39% of its time. Second, the gates this paper set for the sealed seeds do not need the test either. Every pool of every area rose under accept-all, point and matched on all three seeds, the lowest ending at 105% of its start, as under full; the growth gate of 15% in every area was met on two seeds of three by full and matched and on one by accept-all and point. Third, the frequency arm has a side effect. Its prior does not change from one refit to the next, so its repeat proposals fail their tests and the agent's own policy retired the model proposal for 4, 0 and 5 lives on the three seeds; the seed with no retirement still ended 0.184 behind full, so the gap is not that side effect alone, but the arm is not a clean swap of one component. Fourth, the unit is three seeds and each G is one draw of 24 worlds per pool; the next paragraph measures how much that draw matters.

A second evaluation draw. Every number above comes from one draw of 24 worlds per pool. After the arms were read, their 21 retained states were scored again on a second draw that no life, test or check had used, with the script and its salt committed before any state was scored (Table 8). The preregistered verdicts were read on the first draw and stand whatever this one shows; it is reported beside them. Saving every world's result gives each difference from full a paired interval over worlds.

The second draw agrees with the first on every verdict. Matched was ahead of full on all three seeds, by 0.139, 0.125 and 0.082, each with an interval above zero. Accept-all and point were within 0.024 of full on every seed, every interval spanning zero: the same growth, neither less nor more. No-guidance, frequency and no-invention were behind on all three seeds, with intervals below zero.

The draw itself matters more than that agreement suggests. Full's G moved from 0.650, 0.760 and 0.676 on the first draw to 0.737, 0.731 and 0.821 on the second, and single areas moved much further: `drawn7` on seed 919127 read +42% on the first draw and +134% on the second, `drawn11` on 919128 read +66% and +22%. Every area rose again on all three seeds, and `drawn13` on 919127 again missed the 15% gate, at +9%. One pool did not rise. The untrained pool of mechanism worlds on seed 919128 ended at 91% of its start under full, 94% under point, 87% under accept-all and 82% under matched, where the first draw had it rising under all four, by 13% to 32%. "Every pool rose on every seed" is therefore a statement about one draw: on the second, 38 of full's 39 pools rose. On that one pool the two arms that test stayed above the 90% gate and the two that keep everything fell below it, which is one pool on one seed and no more than an observation.

Table 9: Confirmation runs on sealed seeds: tasks solved on fresh held-out sets (24 worlds per pool; 7,200, 2,400 and 4,800 tasks) by the starting and the retained state, the test-to-live cost ratio, and the gate outcome. Version 4 is the confirmed protocol for three areas; version 5 added a guard and was retired; versions 6 to 10 ran six and seven areas (Table 10).

Seed (protocol)	mech	drawn7	metric	cost	gates
919100 (v1)	1,128 to 1,997	236 to 385	405 to 1,055	5.96×	cost fails
919101 (v1)	1,223 to 1,944	271 to 318	318 to 1,619	5.88×	cost fails
919102 (v2)	1,020 to 2,010	281 to 409	364 to 1,550	1.89×	own-set clause fails
919103 (v2)	1,200 to 1,697	255 to 363	363 to 1,428	1.98×	all pass
919104 (v3)	1,276 to 1,807	280 to 364	373 to 1,927	2.29×	one stale baseline
919105 (v3)	1,147 to 2,366	180 to 342	322 to 1,518	1.94×	one stale baseline
919106 (v4)	1,129 to 2,136	202 to 299	343 to 1,647	2.32×	all pass
919107 (v4)	1,138 to 2,263	275 to 382	248 to 1,901	2.04×	all pass
919108 (v5)	1,179 to 1,990	367 to 388	281 to 1,872	2.48×	growth and safety fail
919109 (v5)	1,111 to 2,151	278 to 334	404 to 1,626	2.53×	cost fails

A start with nothing withheld. The starting language lacks two of the nine atom families and twelve of the string positions. That is by design, so that invention has something to find, and every area's worlds use what is withheld. Part of the measured growth is therefore the agent getting back what it was denied. The second draw measures how much: it also scored a start that holds every generator atom and still has no model and no prior. Against the withheld start, that start's G was 0.321, 0.216 and 0.276 on the three seeds, 30% to 44% of full's. About a third of the growth in G is thus available to a start that was simply not handicapped. Most of it lies in the two metric areas, where that start scored 1.6 to 2.7 times the withheld one; in mechanism worlds and string tasks the larger language alone scored below the withheld start, at 79% to 99%, because a larger catalog dilutes a fixed budget. The full agent was ahead of the unhandicapped start by 0.416, 0.516 and 0.546 in G, each interval above zero, and in 20 of the 21 pairs of area and seed; the exception is `drawn7` on seed 919127, at 97%. The growth percentages of Tables 3 and 4 and of the confirmation tables are measured from the withheld start and should be read with this in mind.

The metric areas are small. A metric world is fixed by its centre, and under the two weightings 118 and 80 centres yield a valid world. Full lived 103 to 109 of the first area's and 68 to 70 of the second's on the three arm seeds, so 37 to 47 of the 48 worlds of each first-draw evaluation set, and 39 to 46 of each second-draw set, had a centre the agent had already lived. Evaluation in these two areas, which carry the largest gains in this paper, is closer to a test on worlds seen before than to a test on new ones. Matched, with three and a half times the lives, lived every centre of both areas on every seed, so none of its metric evaluation worlds was new to it, and its lead is partly this coverage. The gains are not confined to lived worlds: on the 33 second-draw worlds whose centre full had never lived it went from 122 tasks to 1,028, and on the 255 whose centre it had lived from 1,837 to 8,737.

7 Interference across areas

A change can improve the area that proposes it while damaging an area the agent has already learned. The first three-area check exposed exactly this failure.

After four lives in metric worlds, the main agent's score on the first drawn partition's untouched held-out set fell from 376 to 338, a 10% loss in an area it had not entered since. The cause was

Table 10: Six- and seven-area confirmation runs (versions 6 to 10; version 8 adds the string-task area, version 9 repairs its worker's consistency filter, version 10 adds the extension test, version 11 the meta-level and 11.1 its full record, and the rest is version 4's code): tasks solved on fresh held-out sets (24 worlds per pool) by the starting and the retained state in each area, the test-to-live cost ratio, and the gate outcome. Version 6's cost gate was 2.5 times living; version 7's is 3.25 at six areas and versions 8 to 10 use 3.5 at seven; for versions 11 and 11.1 the cost is level-0 proofs / all proofs, against 3.5 and 5.0.

Seed (protocol)	mech	drawn7	metric777001	drawn11	drawn13	metric555001	str7	cost, gates
919110 (v6)	1,197 to 2,442	305 to 375	299 to 1,415	291 to 424	375 to 483	356 to 1,833	–	2.78×, cost fails
919111 (v6)	1,187 to 2,160	305 to 384	355 to 1,585	206 to 341	176 to 314	283 to 1,475	–	2.64×, cost fails
919112 (v7)	1,014 to 2,070	214 to 374	493 to 1,723	217 to 415	237 to 375	308 to 1,390	–	2.88×, all pass
919113 (v7)	1,269 to 1,878	230 to 334	389 to 1,667	249 to 369	258 to 375	284 to 1,823	–	2.32×, all pass
919114 (v8)	1,185 to 2,300	166 to 460	344 to 1,422	249 to 381	215 to 439	371 to 1,605	4,749 to 6,026	2.61×, all pass
919115 (v8)	1,174 to 1,885	320 to 413	309 to 1,409	251 to 386	322 to 430	300 to 1,321	4,119 to 5,419	2.71×, all pass
919116 (v9)	1,072 to 2,088	258 to 279	259 to 1,715	220 to 377	224 to 384	339 to 1,485	5,076 to 6,248	2.33×, growth fails
919117 (v9)	1,217 to 2,416	241 to 432	326 to 1,584	275 to 416	305 to 442	384 to 1,716	4,910 to 6,127	2.27×, all pass
919118 (v10)	1,159 to 2,484	273 to 349	450 to 2,283	241 to 439	286 to 379	378 to 1,329	4,722 to 6,193	2.92×, all pass
919120 (v11)	1,164 to 2,185	362 to 463	364 to 1,522	251 to 368	215 to 338	410 to 1,253	4,871 to 6,278	2.95/4.12×, replay fails
919121 (v11.1)	1,309 to 2,356	151 to 326	485 to 1,497	225 to 413	261 to 434	360 to 1,454	4,386 to 6,014	2.10/2.56×, all pass
919122 (v11.1)	1,166 to 2,385	260 to 410	259 to 1,893	303 to 454	214 to 410	458 to 1,764	4,569 to 5,873	2.71/3.67×, all pass

the shared language. The metric lives had accepted one extension of 40 atoms and one removal of 25, each tested on the metric area alone, and a language that grew and shrank to suit metric worlds changed the catalog the drawn partition's model had been fitted over. Each change was an accepted success where it was tested and a regression where it was not.

The revision was to the acceptance rule, made by the experimenter at version 23: a candidate that changes the language is also run on a held-out set of every other area the agent has lived in, 12 worlds per pool, and is rejected if its total there falls more than 2% below the current state's. The current scores of those areas are recomputed after any accepted shared change. The tolerance permits a small measured regression; it is a bound, not preservation. The cost is more held-out lifetimes per language candidate, which is why the combination of all proposals is tested first and the parts only if it fails.

The revised rule fired twice as soon as it existed. In `drawn11` a combination of invented atoms and a model gained 165 to 344 on that area's held-out set and was rejected because it cost mechanism worlds 8% and `drawn7` 12%; the model alone was then accepted, at 417. In `drawn13` a combination worth 150 was rejected for costing `drawn11` 23%. Each would have been accepted under the earlier rule. The `s3` case of Section 6 shows the rule's remaining gap: a 12-world check read a gain where a 24-world set read a 12% loss.

8 The grown language and the pools the agent cannot see

The five-area check at version 24 showed, on the mechanism area's untouched set, that untrained family combinations had fallen from 24 to 18 tasks per world while trained worlds rose from 26 to 50. The acceptance tests had not seen it, because a test compares totals and the untrained pool is a third of the set.

An ablation of the retained state on that set attributes the loss (Table 11). The retained model and prior on the starting language score 1,973 with untrained combinations at 26 per world, above the start; the grown language alone, with no model and no prior, scores 735 against 1,299 for the starting language; the retained state scores 2,072 with untrained at 18. The 166 invented atoms then in the language, 110 of them behaviourally new, had enlarged the catalog from 82,758 rows to

Table 11: Ablation of `s1`'s retained state (version 24) on the mechanism area's untouched held-out set: tasks per world by pool, and total of 7,200.

State	Trained	Unfamiliar	Untrained	Total
retained	49.5	18.7	18.2	2,072
retained, no prior	43.0	14.7	9.5	1,614
retained, no model	37.3	15.0	15.9	1,639
grown language only, no model, no prior	11.1	9.0	10.5	735
starting language, retained model and prior	50.0	6.1	26.2	1,973
starting state	26.2	3.7	24.2	1,299

266,272, and the half of each task's budget that the prior rule reserves for the no-prior order was scanning a catalog two thirds of which the agent had added. Trained worlds did not suffer, because the model's half of the budget found their rows anyway; untrained combinations, which the model promotes less, did.

The agent already held the verdict it needed. Its self model files an atom in no group when the agent would almost never find it in a world. Across the five area models, 107 invented atoms were rated real by at least one model, 55 of the 56 that reproduce a withheld family among them, and 59 were rated import-only by every model that had seen them, one of them real. From version 26 the pruning operation proposes those too, and the acceptance test decides.

Two further revisions went in at the same version, both to the test. The interval requirement replaced the strict-inequality rule after the life-10 case. And the tie-break, the order the scan uses among rows it scores equally, had been a random permutation of row indices, so adding or removing any atom reshuffled every other row and every language test carried that noise; it is now a hash of each row's content, so a language change changes nothing but the rows it adds or removes.

On life 11 the removal of the 55 atoms was accepted, +0.3 to +3.0 tasks per world on the mechanism set and better in all four other areas, and of the model and extension proposed beside it, the extension was rejected on an interval below zero and the model on one that crossed zero. The language went from 426 to 371 atoms. On the untouched set untrained combinations now read 18.6 against a start of 20.5, inside the 10% that the draft confirmation protocol allows, with every other pool and area above its start. Most of that narrowing is the baseline and not the removal: under the earlier tie-break the starting state read 24.2 on this pool and the retained state 18.2 (Table 11), so the new tie-break lowered the start by 3.7 tasks per world while the removal raised the retained state by 0.4.

9 What this does and does not show

Table 12 states each claim a reader might take from this paper, whether the evidence supports it, and what limits that evidence. The paragraphs after it give the detail, and the section ends with the list of limitations.

It shows one persistent agent improving its own state in five areas of different structure, and two further agent instances improving in three areas and two, keeping only changes it tested on worlds it never trained on, with the tests charged. The three forms of growth from the antecedents appear together in its ledger: within-life accumulation in every life, learned search guidance accepted nearly every life, and invented primitives that let it reach families its starting language could not express. The preregistered arms separate the last two: without guidance the agent reached 6% of the full

Table 12: Claims, their status, and the evidence for each.

Claim	Status	Evidence and its limit
The agent's state improves on held-out sets under the final protocol	Supported	Ten sealed seeds: 70 of 70 area scores rose and nine seeds passed the four gates, seven of the eight frozen in advance. Seven fixed area structures, one platform, one draw per seed; on a second draw of three seeds every area rose again and 38 of 39 pools.
Learned guidance carries the growth	Supported on three seeds	Without it 6% of the full agent's growth, lower on every seed (preregistered rule 4); behind on every seed of the second draw, intervals below zero.
Invention carries growth where the starting language cannot express the rules	Supported on three seeds	Without it 65% of the growth and fewer tasks in the unfamiliar pools on every seed, about half in mechanism worlds (rule 4); behind on every seed of the second draw.
The acceptance test is what makes the agent grow	Withdrawn	Keeping every change grew as much for 29% of the scans, and spending the test's scans on experience grew more on three of three seeds (rule 2), on both draws.
The acceptance test keeps out harmful changes	Suggestive	2 of 35 kept changes harmful with the test, 15 of 63 and 17 of 57 without. Raw counts, a noisy label, three seeds.
The agent improves its own procedure	Narrow	13 adoptions from a menu of six settings. Twelve came from tests in which nothing differed and saved nothing over a whole run; one of ten differing tests gave a better state, unreplicated.
The growth is more than recovering the withheld primitives	Supported on three seeds	A start with nothing withheld reaches 30% to 44% of the full agent's G; the full agent is ahead of it on every seed, interval above zero.
Recursive self-improvement in the open-ended sense	Not shown, not claimed	The agent chooses among settings it was given and writes no new procedure.
The method works in a practical domain	Not tested	Every area is a synthetic rule-inference task.
The runs can be reproduced exactly	On the reference platform	Every accepted change replays there and one run was re-executed end to end; on Linux two of seven areas draw different worlds.

agent's growth and without invention 65%, on every seed. Growth in one area did not require starting over in another; the language carried, and in the metric area a model fitted from 48 worlds was worth 975 tasks at once.

It does not show that testing changes is what makes the agent grow. At a budget of 1,024 scans and on these seven areas, keeping every proposal grew as much as testing them for less than a third of the scans, and spending the test's scans on more experience instead grew more on all three seeds. A second evaluation draw gave the same reading on every seed. The test's measured benefit is safety: it kept far fewer changes that hurt their own area. A reader who wants growth per scan from this design need not test at all; a reader who wants a carried state that rarely gets worse should. Whether the balance shifts with more areas, a tighter budget or longer runs is open.

The development results of Table 3 were inspected as the design changed, on random seeds used throughout. The confirmation runs (Table 9) are the prespecified evidence: a frozen design, pre-stated gates, a committed forecast, thirty sealed seeds in twelve versions, and a replay of every accepted change. The cost gate failed in the first version; the audit found one bookkeeping flaw in each of the second and third versions, each fixed before the next pair of seeds; the fourth version passed all four gates on both seeds; the fifth added a guard, failed growth and safety on one seed and cost on the other, and was retired; the sixth and seventh widened the schedule to six areas, three of them new to every seed, and the seventh passed all four gates on both seeds; the eighth added a

seventh area of a second task form and passed all four on both seeds; the ninth repaired the string worker and failed growth on one seed at the gate's noise floor; the tenth added the extension test and passed on its first seed. The eleventh added the meta-level and failed replay on one meta-test of its one seed; 11.1 completed the record, and on ten sealed seeds nine passed gates 1 to 4. No replay overturned an acceptance. Growth of at least 15% in every area held on seventeen seeds of nineteen, and no pool below 90% of its start on eighteen of nineteen; the exceptions are seed 919108 under the retired version and, for growth, seed 919116's drawn partition at +8%. Across the nine seeds of six or more areas, 59 of 59 area scores rose, by 8% to 562%, and 105 of 105 pools rose. The held-out sets behind both tables are single draws whose per-world results were not saved, so no interval accompanies the totals. Two accepted changes of a development seed's chain run do not replay, for a reason not found; the auditor replayed the 171 accepted changes of the nineteen confirmation seeds and found the two stale baselines above and nothing else.

Through version 10 it is self-improvement of state, not of procedure. The agent's model and language direct its search, its search produces the experience the next model and language are fitted from, and the ledger shows the state improving the state across twelve lives. But the procedures that turn experience into a model, candidates into atoms, and tests into acceptances are fixed code, and the revisions of the acceptance rule were made by the experimenter.

Version 11.1 moves one step up, on ten sealed seeds. The agent changes settings of its own improver, from evidence in its own ledger, under a test no proof used, and runs its later lives under the adopted settings; every such decision replays. That is a change to the procedure, made by the agent, under proof. It is a small one. The menu is six settings of two or three values each, written by the experimenter; the agent chooses among procedures it was given and does not write new ones. Twelve of the thirteen adoptions made the improver cheaper, not better, seven of the thirteen were the same setting, and the one better adoption is a single test whose interval's lower end is 0.1 tasks per world. Nor can an adopted setting change how the agent proposes or judges mutations: the meta-level itself is fixed. The programme's definition, a successor with a better improvement procedure, is met in this narrow sense and not in the open-ended one.

The growth also saturates and is uneven. On the main agent's later mechanism lives few candidates passed, and model gains fell from hundreds to single digits by the sixth life. On the main development agent untrained combinations in mechanism worlds ended 9% below their start, a loss that drove later revisions; on the ten seeds of the final protocol that pool rose on every seed, by at least 13%. Half of the remaining invented atoms are behaviourally new and some are surely junk that costs budget until a removal takes them out. Testing cost two and a half times living. And every area is a synthetic rule-inference task, in two task forms; the agent has not met a practical domain. A reviewer suggested a language-model coding agent that accumulates reusable functions under the same test. That would be a different system, since this agent has no language model and none of its mechanisms was designed for one, and it is outside this paper.

Limitations. An audit of this paper against its own repository, run before posting, produced the following list. Each item is a way the results above could mislead, with the measurement that bounds it where one exists.

- *The starting state is handicapped by design.* It lacks two atom families and twelve string positions that every area uses. A start with nothing withheld reaches 30% to 44% of the full agent's growth in G before anything is learned, most of it in the metric areas; the full agent is ahead of that start on every arm seed (Section 6). The other seven sealed seeds were not scored against it.
- *One evaluation draw.* The ten-seed results, the gates and the harm counts each rest on one draw of 24 or 12 worlds per pool. A second draw of the three arm seeds confirmed every arm verdict

and moved single areas by tens of percentage points; one pool that rose on the first draw fell to 91% of its start on the second (Section 6). The other seven sealed seeds have one draw only.

- *Seeds vary worlds, not areas.* All ten seeds meet the same three drawn partitions and two metric weightings (Section 4). The intervals of Table 4 are over random seeds for those structures.
- *The metric areas are small.* They hold 118 and 80 distinct worlds, and 77% to 98% of the metric worlds in an evaluation set had a centre the agent had lived; gains on never-lived centres are as large (Section 6).
- *The harm count is a noisy label.* A kept change counts as harmful when its area's total fell on one draw of 12 worlds per pool. Many such drops are within that draw's noise, the unit is three seeds, and no significance is claimed for the difference between 2 of 35 and 15 of 63.
- *Matched is matched on scans only*, and took 2.5 times full's wall-clock time (Section 6).
- *Some cross-area sets were not fresh.* A cross-area check draws the other area's set with the proposing area's salt. On the ten seeds, 34 of 249 such checks used a salt that an earlier test in the checked area had used, so the first pool's worlds, and for a single-pool area the whole set, had already informed a decision there. Salts should be derived per purpose.
- *The protocol was revised between versions.* Seven of eleven revisions followed a failed gate and four did not. The cost gate's threshold was moved once after a version had been read (at version 7; version 6's seeds are counted as failures), and the limit of 5.0 on all tests was set at version 11 after a development run read 3.52.
- *One platform.* Exact replay holds on the reference platform. On Linux with the same versions the two metric areas draw different worlds (Section 13), so a rerun there is a statistical replication. A replay re-scores accepted changes and does not re-execute a run; one run was re-executed end to end and matched.
- *The hashes cover eleven files*, not every module they import, and one sealed seed (919119) was interrupted and left no data.
- *Procedure changes.* Twelve of thirteen adoptions came from tests in which nothing differed and saved nothing over a whole run; the thirteenth is one pass among ten differing tests (Section 6).

10 Related work

Continual learning. Continual learning asks a learner to keep performance across a sequence of tasks without forgetting [13, 9]. The interference of Section 7 is an instance of the same problem. The remedy differs: continual-learning methods usually constrain a training update, while here a discrete change to shared state is run on every area it touches and refused if any falls beyond a tolerance. Nothing here is trained by gradient, and the tolerance bounds a measured loss rather than preventing one.

Adaptive search. Learned guidance for enumerative program search goes back at least to DeepCoder [1] and RobustFill [3], which predict from examples what a program contains and order the search accordingly. The guidance here is of that kind, but it is fitted only from what the agent earned while solving, with no correctness labels and no synthetic training corpus. The half-set stop and the extension set resemble the adaptive resource allocation of hyperparameter search [10], applied to a go or no-go decision about one change.

Program synthesis and library learning. DreamCoder [5] is the closest antecedent: it alternates solving tasks, growing a library from solved programs, and training a search policy, so that library and policy improve each other. Three things differ here. Primitives are proposed from what recurs in tasks the agent could not solve, rather than compressed out of programs it did solve. Every addition and removal has to pass a paired held-out test, and a change to the shared language has to pass one in every other area. And the language here is far narrower than DreamCoder's, with seven small areas rather than domains such as planning and inverse graphics. Later library learners compress or refactor solved programs with a faster search or with a language model (Stitch [2], LILO [6], ReGAL [17]); all of them grow the library from solutions, where the agent here proposes primitives from failures.

Self-improving systems. The Gödel machine [16] rewrites itself only on a formal proof that the rewrite is useful. Later systems replace the proof with measurement: the Darwin Gödel Machine [23] has coding agents modify their own code and validates each change on benchmarks, STOP [22] applies a language-model improver to its own code, and AutoML-Zero [14] evolves learning algorithms from basic operations. Among language-model agents, Meta Agent Search [7] has a meta agent write new agents into a growing archive, the Gödel Agent [21] lets a model rewrite its own logic, and the Huxley-Gödel Machine [20] guides self-modification by an estimate of how a modification's descendants will perform. All of these change much more of the procedure than the agent here does, whose procedure changes are confined to a fixed menu. What this paper adds is on the side of evidence: pre-stated gates, sealed seeds, charged test costs, a replay of every decision, and preregistered ablations of its own parts.

Keeping a change only on evidence. The acceptance rule has older and closer relatives, and none of its ingredients is new here. The success-story algorithm [15] keeps a learner's self-modification only while reward per unit time since the modification exceeds what came before it, and charges the time spent: a test-then-keep rule with its cost accounted, in reinforcement learning. High-confidence policy improvement [19] returns a new policy only when a lower confidence bound on its performance, computed on held-out data, exceeds the current policy's; the interval-above-zero rule used here is of that family, without its guarantee. Among agents that edit a library of skills, GRASP [12] admits an edit only on a net improvement on a balanced held-out probe under a regression budget, which is close to this paper's own-area test combined with its cross-area tolerance, and Ratchet [24] studies when a skill can be retired, the counterpart of pruning. Gradient episodic memory [11] constrains each update so that the loss on stored examples of earlier tasks does not rise, the continual-learning analogue of the cross-area check. Population-based training [8] forks learners, compares them and copies the better one's settings, as the meta-level does with one fork. And drawing a new held-out set after every acceptance is a blunt answer to the adaptive reuse of a holdout that [4] treat properly. What this paper adds to that line is a measurement on its own system: under rules fixed in advance, the test bought safety and not growth.

11 Lessons

1. Guidance and invention are the engine; the acceptance test is the brake. Without either of the first two the growth fell on every seed, to 6% and 65% of the full agent's. Keeping every proposal grew as much as testing them for less than a third of the scans; what the test bought was 2 harmful changes kept in 35 against 15 in 63. Measure what a test is for before calling it the design.

2. Test the combination first, then the parts. This halved the tests in the common case; whether it loses anything against testing every part was not measured.

3. Invention needs recurrence. An atom that fits one uncovered task is usually a coincidence; one that fits two is usually a missing atom. The first version accepted single fits, and every extension it proposed lost.

4. Let the agent remove what it added. Removing atoms that never earned their keep was accepted three times in four, and removing the atoms the self model called import-only was accepted with a gain in all five areas. It did not by itself undo the loss on untrained combinations, most of whose apparent recovery was a change of baseline (Section 8).

5. A model from 48 worlds can be worth keeping, when a test decides.

6. Shared state needs shared tests. The 10% loss in an untouched area was invisible to any single-area test.

7. Score growth on worlds the agent never saw, and attach an interval. A gain of 24 on 72 worlds is noise.

8. Make the measurement stable under the change being measured. A tie-break over row indices turned every language change into a reshuffle of the whole catalog.

9. Build for the host. One stall and one crash in the record were the machine's shared memory, not the agent.

12 Next

- Run the arms on more seeds, with intervals, and at other budgets and area counts, to find where, if anywhere, the test pays for itself in growth and not only in safety.

- A cheaper safety check: the test's benefit is fewer harmful changes, and a check aimed at harm alone may cost far less than a full paired test.

- Replicate a procedure change that yields a better state and not only a cheaper one; on the ten seeds of the final protocol there is one.

- Save per-world results in the final checks, as the second draw does, so that each seed's reported gain carries its own interval and not only the interval over seeds; and score every sealed seed on more than one draw.

- Repair what the audit found in the design: report growth against a start with nothing withheld as well; withhold a share of the metric centres from every life; derive each held-out set's salt from its purpose; and order equal distances in the metric areas by a stable sort, so that a run replays across platforms.

- The cross-area check is the tightest part of the test: 2% admits a third of the candidates that reach it. Its size and tolerance need setting against the harm they prevent.

- A guard against cumulative drift in the pools the agent cannot see, compared with the starting language's unguided search, if the untrained-combination loss returns.

- A practical domain, with the agent's solver as one of several.
- Past the fixed menu: let the agent propose new procedure components (a new vote rule, a new fallback order, a held-out set size per kind of candidate) rather than step through given values, under the same meta-test; and replicate the better-not-cheaper adoption, which rests on one test.

13 Reproducibility

The repository is `https://github.com/MMWilliams/rsi`. This paper corresponds to the tag `seed-growth-arxiv-v1`, and the arms' preregistration is the commit tagged `prereg-review002-arms`.

Without running the agent. `shreproduce_tables.sh` checks the hashes of the frozen files, regenerates every analysis file and every generated table of this paper (Tables 4, 5, 6, 7 and 8) from the committed states, ledgers and logs, and fails if anything differs from the committed copy. It needs Python and NumPy and takes about two minutes. Every number in Section 9's limitations comes from `analysis/review002-limits.json`, `analysis/review002-second-draw.json` or `analysis/review002-metric-coverage.json`, which the same script rebuilds, except the last, which draws worlds and so needs the agent's environment.

Environment. The agent needs Python 3.12.9, the packages pinned in `requirements-seed.txt`, and the string language of [18]. That language is a separate repository that carries no licence, so it is not redistributed here: clone `https://github.com/izzortsi/trapdoor-loom` into `vendor/trapdoor-loom` at commit `dd78141`. `platform_check.py` then reports in about 20 seconds whether the machine draws the same worlds as the reference platform.

Re-running a seed. `shconfirm_seed.sh<newname><randomseed>` runs the frozen schedule, the final check and the audit from scratch. It must be given a new name: the same command on an existing name resumes that agent and extends it. The result is compared with the committed `seeds/c22` to `seeds/c31`. `run_arms.py<arm><tag><randomseed>` does the same for an arm, and `second_draw.pyscore` and `metric_coverage.py` redo the second draw and the count of metric worlds. The lives of the full arm take about 37 minutes with six worker processes on a 12-core desktop processor (Ryzen 9 9900X, 64 GB), and the sealed schedule adds its meta-tests to that.

What the records cover. The hashes in `v11.1.sha256` and `arms.sha256` cover the eleven files that hold the agent, its runners, the final check and the audit. They do not cover the other modules those files import, the string language or the library versions; the tag and the pinned requirements fix those. A replay by `audit_seed.py` re-scores each accepted change, policy revision and meta-test from its saved snapshot and compares the result with the ledger. It does not re-execute lives, rejected candidates, proposals or cross-area checks. One run does re-execute all of those: seed 919127 adopted no procedure change, so its full arm, run from scratch two weeks after the sealed run and with a different number of workers, repeats that run, and all 78 of its live and test records, its language, its kept changes, its experience and its final check equal the sealed run's. The preregistration of the ten-seed replication (commit `74d1628`) was pushed before its runs. The arms' preregistration (`015ef03`) and the second draw's script (`5dd2d33`) were committed locally before their runs and pushed after, so their timing rests on local commit dates.

Files. The seed agent is `rsi/seed.py` (state, areas, the four operations, the acceptance test) and `rsi/seed_string.py` (the string-task form) with `rsi/seed_areas.py` for metric worlds, `rsi/invention.py` for the grammar and the recurrence test, and `rsi/self_world_model.py` for the model. `seed_run.py--names1--seed644101--areamech--lives4` runs lives and resumes from the saved state; `seed_check.py--names1--areasmech,drawn7,metric777001` scores an agent on held-out sets no test used and labels its inventions; `seed_ablate.py` produced Table 11 from the version-24 state under the earlier tie-break, so it regenerates at commit `ffaa39d` and not at the head; `audit_seed.py` replays accepted tests from version snapshots; `confirm_seed.sh<name><seed>` runs the frozen confirmation schedule, check and audit of `EXPERIMENT-seed-confirmation.md`, whose runs are in `seeds/c1` to `seeds/c31`, four over six areas and the rest over seven in two task forms (there is no `seeds/c20`: seed 919119, the second version 10 seed, was interrupted, never rerun and left no data). The worked task of Section 2 is reproduced by drawing the first trained world of `MechanismArea(644101)` with `random.Random(7)` and the task from its first mechanism with `random.Random(11)`. States are in `seeds/<name>/state.json`, committed after every run; the ledger inside each is the full record of every candidate, its test and its cost. Logs are `runs_seed_*.log` and the checks `analysis/seed-check-<name>.json`. `summarize_v11_replication.py` and `rejudge_proofs.py` produce the analysis files behind the ten-seed and re-judging tables. The arms are `rsi/seed_arms.py` and `run_arms.py`, run one at a time by `run_review_arms.sh`, with their file hashes in `arms.sha256`; their runs are in `seeds/<arm>_c24` to `seeds/<arm>_c26`, and `summarize_arms.py` scores them against the rules. `summarize_limits.py` counts what Section 9's limitations cite. `paper_tables.py` writes the ten-seed, re-judging, arms and second-draw tables from the analysis files. The two figures and the hand-typed tables have no generating script; their numbers are in `analysis/seed-check-<name>.json`.

Every run reported here was made on Windows 11 with Python 3.12.9 and NumPy 2.5.3. The same versions on Linux, on the same processor, draw different worlds in the two metric areas, because the last bits of `numpy.fft.rfft` differ and `numpy.argsort` breaks ties among equal distances differently (`rsi/seed_areas.py`); the other five areas draw the same worlds. Reproducing the metric-area numbers exactly needs the reference platform.

Settings of the development agents (the final protocol's are in Section 5): 48 worlds of 100 tasks per life; budget 1,024 scans per task; 24 worlds per pool for an area's held-out set and 12 for the cross-area check; tolerance 2% relative; group counts 6 to 8 and 2 to 4 groups per world by held-out likelihood; extension cap 40 atoms per life with recurrence 2; removal after 2 lives or on the self model's import-only verdict; tie-break a content hash per row with seed 77; a change accepted only when the paired 95% bootstrap interval (4,000 draws) of its gain is above zero. Area specifications are stored in the state so that a test can rebuild any area the agent has lived in. `RSI_WORKERS` caps the worker pool; six or fewer is safe on this machine beside other jobs.

Agents: `s1` on random seed 644101 (versions 1 to 26 reported here), `s2` on 643101 and `s3` on 645101. The companion paper on the world and self model is `SELF-SIMULATION-WORLD-MODEL-2026-10-10.md`, and the distillation of earlier growth results is `GROWTH-DISTILLATION-2026-10-10.md`.

Use of AI tools. The code, the experiment scripts, the analysis and the drafts of this text were written with an AI coding assistant (Claude, by Anthropic) working under the author's direction, and the audit that produced the limitations of Section 9 was run with the same assistant. No language model is part of the agent studied. The author is responsible for the content.

References

[1] M. Balog, A. L. Gaunt, M. Brockschmidt, S. Nowozin and D. Tarlow. DeepCoder: Learning to Write Programs. ICLR, 2017.

[2] M. Bowers, T. X. Olausson, L. Wong, G. Grand, J. B. Tenenbaum, K. Ellis and A. Solar-Lezama. Top-Down Synthesis for Library Learning. POPL, 2023.

[3] J. Devlin, J. Uesato, S. Bhupatiraju, R. Singh, A. Mohamed and P. Kohli. RobustFill: Neural Program Learning under Noisy I/O. ICML, 2017.

[4] C. Dwork, V. Feldman, M. Hardt, T. Pitassi, O. Reingold and A. Roth. The Reusable Holdout: Preserving Validity in Adaptive Data Analysis. Science 349(6248), 2015.

[5] K. Ellis, C. Wong, M. Nye, M. Sablé-Meyer, L. Morales, L. Hewitt, L. Cary, A. Solar-Lezama and J. B. Tenenbaum. DreamCoder: Bootstrapping Inductive Program Synthesis with Wake-Sleep Library Learning. PLDI, 2021.

[6] G. Grand, L. Wong, M. Bowers, T. X. Olausson, M. Liu, J. B. Tenenbaum and J. Andreas. LILO: Learning Interpretable Libraries by Compressing and Documenting Code. ICLR, 2024.

[7] S. Hu, C. Lu and J. Clune. Automated Design of Agentic Systems. ICLR, 2025.

[8] M. Jaderberg, V. Dalibard, S. Osindero, W. M. Czarnecki, J. Donahue, A. Razavi, O. Vinyals, T. Green, I. Dunning, K. Simonyan, C. Fernando and K. Kavukcuoglu. Population Based Training of Neural Networks. arXiv:1711.09846, 2017.

[9] J. Kirkpatrick et al. Overcoming Catastrophic Forgetting in Neural Networks. PNAS 114(13), 2017.

[10] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh and A. Talwalkar. Hyperband: A Novel Bandit-Based Approach to Hyperparameter Optimization. JMLR 18(185), 2018.

[11] D. Lopez-Paz and M. Ranzato. Gradient Episodic Memory for Continual Learning. NeurIPS, 2017.

[12] J. Moll, J.-P. Corbeil, J. Pan, M. Hadamitzky, D. Rueckert, L. Adams and K. Bressem. GRASP: Gated Regression-Aware Skill Proposer for Self-Improving LLM Agents. arXiv:2605.29668, 2026.

[13] G. I. Parisi, R. Kemker, J. L. Part, C. Kanan and S. Wermter. Continual Lifelong Learning with Neural Networks: A Review. Neural Networks 113, 2019.

[14] E. Real, C. Liang, D. So and Q. Le. AutoML-Zero: Evolving Machine Learning Algorithms From Scratch. ICML, 2020.

[15] J. Schmidhuber, J. Zhao and M. Wiering. Shifting Inductive Bias with Success-Story Algorithm, Adaptive Levin Search, and Incremental Self-Improvement. Machine Learning 28, 1997.

[16] J. Schmidhuber. Gödel Machines: Fully Self-Referential Optimal Universal Self-Improvers. In Artificial General Intelligence, Springer, 2007.

[17] E. Stengel-Eskin, A. Prasad and M. Bansal. ReGAL: Refactoring Programs to Discover Generalizable Abstractions. ICML, 2024.

[18] I. L. Strozzi. Teacher-Free Self-Training Amplifies but Does Not Compound: A Pass@K Crossover on a Free-Verifier Domain. arXiv:2606.07856, 2026.

[19] P. S. Thomas, G. Theocharous and M. Ghavamzadeh. High Confidence Policy Improvement. ICML, 2015.

[20] W. Wang et al. Huxley-Gödel Machine: Human-Level Coding Agent Development by an Approximation of the Optimal Self-Improving Machine. arXiv:2510.21614, 2025.

[21] X. Yin, X. Wang, L. Pan, L. Lin, X. Wan and W. Y. Wang. Gödel Agent: A Self-Referential Agent Framework for Recursive Self-Improvement. ACL, 2025.

[22] E. Zelikman, E. Lorch, L. Mackey and A. T. Kalai. Self-Taught Optimizer (STOP): Recursively Self-Improving Code Generation. COLM, 2024.

[23] J. Zhang, S. Hu, C. Lu, R. Lange and J. Clune. Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents. ICLR, 2026.

[24] X. Zhang, Y. Cui, G. Wang, Z. Li, W. Qiu, B. Zhu and P. He. Ratchet: How Reliable Must an LLM Judge Be to Retire a Skill? arXiv:2605.22148, 2026.

A Chronology of the protocol versions

After the design was frozen, with the hashes of its source files, a protocol with four pre-stated gates and a committed forecast was run on random seeds never used before: four mechanism lives, three in `drawn7`, three in metric worlds, two more mechanism lives, then the fresh check of Table 3 and a replay of every accepted change. The gates were growth of at least 15% in every area, no pool of any area below 90% of its start, testing at most 2.5 times the cost of living, and every accepted change replaying exactly from its own held-out set. The protocol went through four versions, each frozen with new hashes and run on two new sealed seeds, and each revision came from a failed gate (Table 9). Version 1 passed growth, safety and replay on both seeds and failed cost on both, at 5.96 and 5.88 times living; the arithmetic is that a fresh held-out set every life makes the re-score of the current state alone 1.5 times a 48-world life in the mechanism area, with three candidates per life on top. Version 2 changed the schedule's economics and not the acceptance rule: lives of 96 worlds, 48-world sets in every area, a held-out set kept until something is accepted on it, and a stop at the half set when the mean gain is not positive. It passed cost on both seeds, at 1.89 and 1.98, and failed on one of them the clause that each acceptance has its own set, because two parts of one life were accepted on the same set. Version 3 drew a fresh set after any acceptance within a life and failed the replay once on each seed, by 47 and 27 tasks in the recorded baseline: the current state's cached score on a reused set had gone stale as the area's experience, from which its search prior is built, grew. Every acceptance stood on replay. Version 4 keys that cache on the experience count as well as the version and passed all four gates on both of its seeds, at 2.32 and 2.04 times living. Across the eight seeds every area rose by at least 17% and every pool of every area rose except one, which ended at 94% of its start. The cross-area check rejected shared changes on most seeds for losses on the mechanism area that the proposing area's own test had accepted. Version 5 added the guard the development results had asked for: a candidate that adds atoms to the shared language is also scored with the model and prior switched off, on half the set, and rejected if its unguided search falls more than 10% below the carried language's. On seed 919108 the guard never fired, three extensions passed at unguided losses of 2% to 9% each, and the untrained pool ended at 89% of its start while the drawn partition grew 6%; on seed 919109 the guard fired three times, every

pool rose, and testing cost 2.53 times living. A per-candidate tolerance does not bound a cumulative drift, and the guard was retired; version 4 is the confirmed protocol for three areas. Versions 6 and 7 kept version 4's code byte for byte and widened the schedule to six areas: the three above, two further drawn environments (`drawn11`, twelve groups with four per world and no slip; `drawn13`, six groups with two per world and slip 0.15) and a second metric weighting (`metric555001`), none of the three lived by any seed before, in 18 lives (Table 10). Version 6 passed growth, safety and replay on both seeds, every one of the twelve area scores and twenty pools rising, and failed cost on both, at 2.78 and 2.64 times living: a shared-language candidate that passes its own area is then checked on every other lived area, so the test bill grows with the number of areas, by 0.18 per area over version 4's mean. Version 7 changed only the cost threshold, to 2.5 plus 0.25 per area beyond three (3.25 at six), the 0.25 being the structural ceiling of one 24-world check against a 96-world life per candidate; it passed all four gates on both seeds, at 2.88 and 2.32 times living. Version 8 added the string-task area (`str7`), a second task form over a catalog of its own: version 7's schedule plus three string lives, 21 lives in all, with the cost threshold at 3.5. It passed all four gates on both seeds: fourteen of fourteen area scores rose, by 27% to 356% with a median of 78%, 26 of 26 pools rose, 24 of 24 acceptances replayed, and testing cost 2.61 and 2.71 times living, inside the six-area range, because a form whose catalog is disjoint adds its own tests and no cross-form check. The string area rose 27% and 32%, and both seeds kept the same six withheld positions, the end of the first, second and third upper-case run and the start of the first, second and third lower-case run, proposed from what stuck worlds had earned and accepted on held-out tasks with no label read; one seed also kept two positions outside the generator's universe. Version 9 changed the string worker alone: its consistency test had excluded any row that returns an empty string on some visible input, which the generator's own rows do when two positions coincide, so that 283 of 2,400 development tasks had no expressible program at any budget; with a row allowed to be empty on some inputs and not all, 16 remained, and the same carried state solved 2,015 of those tasks instead of 1,901. On the sealed seeds the string area started higher, at 5,076 and 4,910, and still rose 23% and 25%, eleven of the twelve withheld positions invented on each; but seed 919116 failed the growth gate in `drawn7`, at +8%, an area the change did not touch. A replay of that area's fresh set after each accepted change showed no drift: it stood at +10% after the area's own accepted change, later shared-language changes moved it by at most 24 tasks, and the area's remaining candidates, at +7% to +10% on 48 worlds, had been rejected on intervals that straddled zero. Version 10 therefore runs a candidate whose full-set interval straddles zero, with a mean gain large enough that a doubled set would clear it, on one extension set of the same size and decides on the union, and the auditor replays both sets. Its first sealed seed passed all four gates at 2.92 times living, two of its fifteen acceptances made on an extension set and both confirmed there at a similar gain; the second, seed 919119, was interrupted by the machine at the twentieth of 21 lives, was not rerun, and left no data. Version 11 added the meta-level above to version 10's schedule, with meta-tests every second life and a fifth gate, at least one adopted mutation, reported but not required; the cost gate became 3.5 times living for level-0 proofs and 5.0 for all proofs. Its seed 919120 passed growth, safety, cost and the fifth gate, adopting a recurrence of 3 and a pruning threshold of 0.05, each the same state at 15% to 17% less proof cost, and rejecting a patience of 2 that would have cost 42% less but scored 730 to 683 on its meta set. It failed replay on that rejection: the auditor scored the losing branch at 699, because over a two-life horizon the loser lives its second life under a different state, and its experience from that life had not been recorded. Re-executing that life from the loser's snapshot reproduced the record world for world; the decision was right and the record incomplete. Version 11.1 records the loser's later experience and changes nothing the agent decides. Both of its sealed seeds passed all five gates (Table 10): every area rose, by 29% to 631%, every pool rose, level-0 proofs cost 2.10 and 2.71 times living and all proofs 2.56 and 3.67, and every acceptance, policy

revision and meta-proof replayed. Of eight mutations tested, four were adopted: an extension cap of 80 on both seeds and a pruning threshold of 0.05 on one, each the same state for 15% or 16% less, and on 919122 a recurrence of 3 that produced a better state, 724 to 759 on its meta set (+0.1 to +1.4 tasks per world) at 36% less proof cost (17.2 million scans against 27.0 million; the other branch needed an extension set). The other four left the state unchanged at equal cost and were declined. Forecasts were committed before each version ran; the probability given to all four gates passing on both seeds was 0.15, 0.35, 0.4, 0.3, 0.15, 0.2, 0.35, 0.3, 0.35, 0.3, 0.35 and 0.35.